

Wireless Access Electromagnetics

LAB 4

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Part 1:

- This report includes the analysis of a patch antenna using microwave CST in 2 cases;
 - Ideal substrate
 - Real substrate
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- I assessed the S11 parameter over the 5.3–6.3 GHz frequency range and illustrated the far-field radiation patterns in the E and H plane at the center frequency.

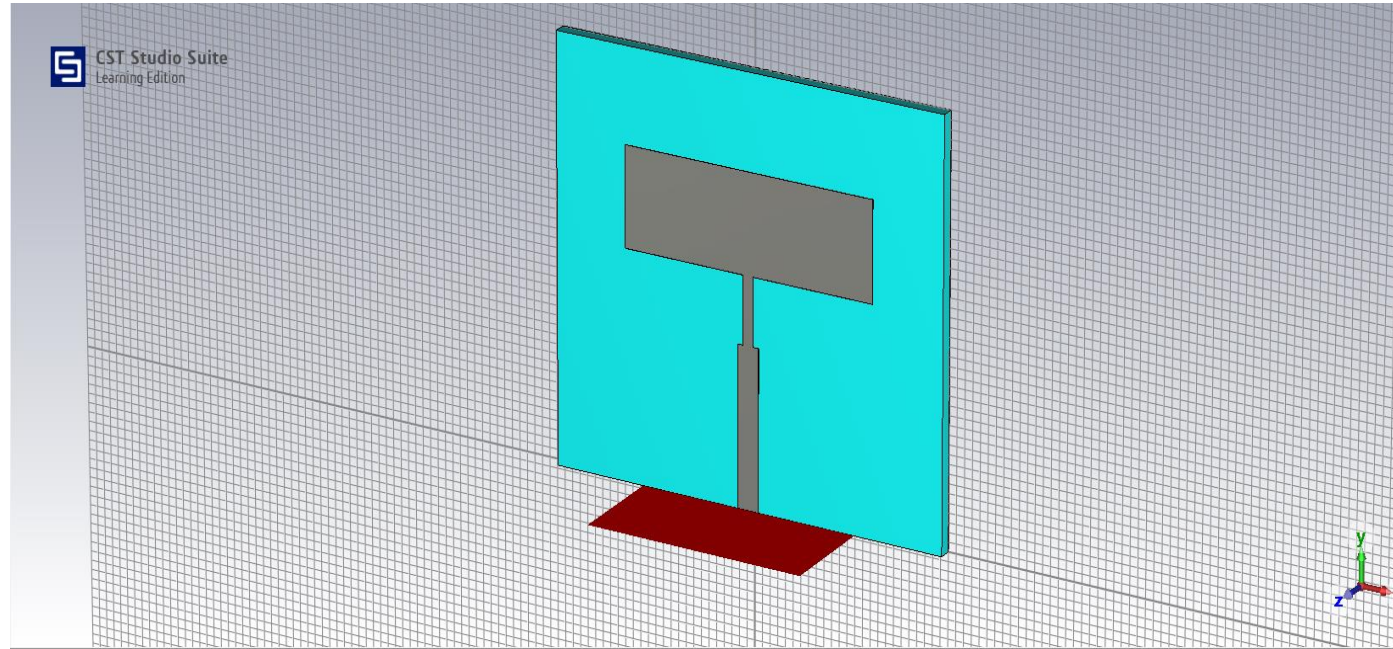


Figure: The CST geometry

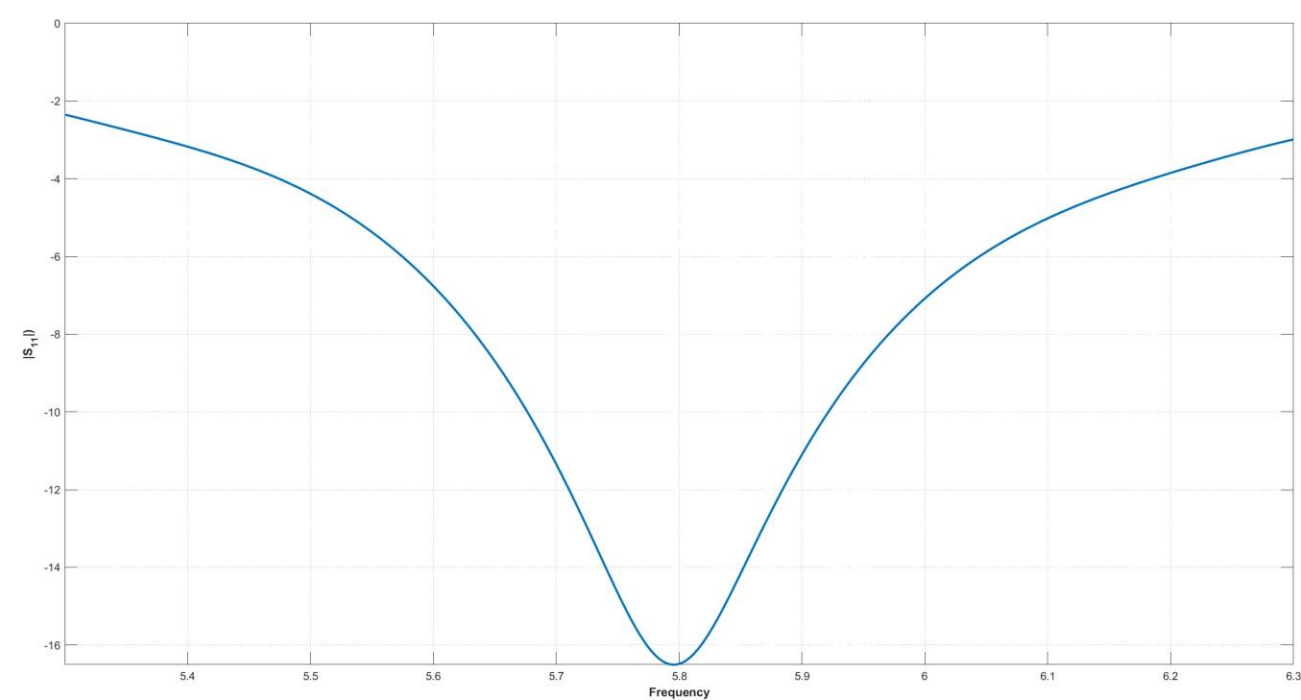


Fig 1: S11 – real substrate with losses

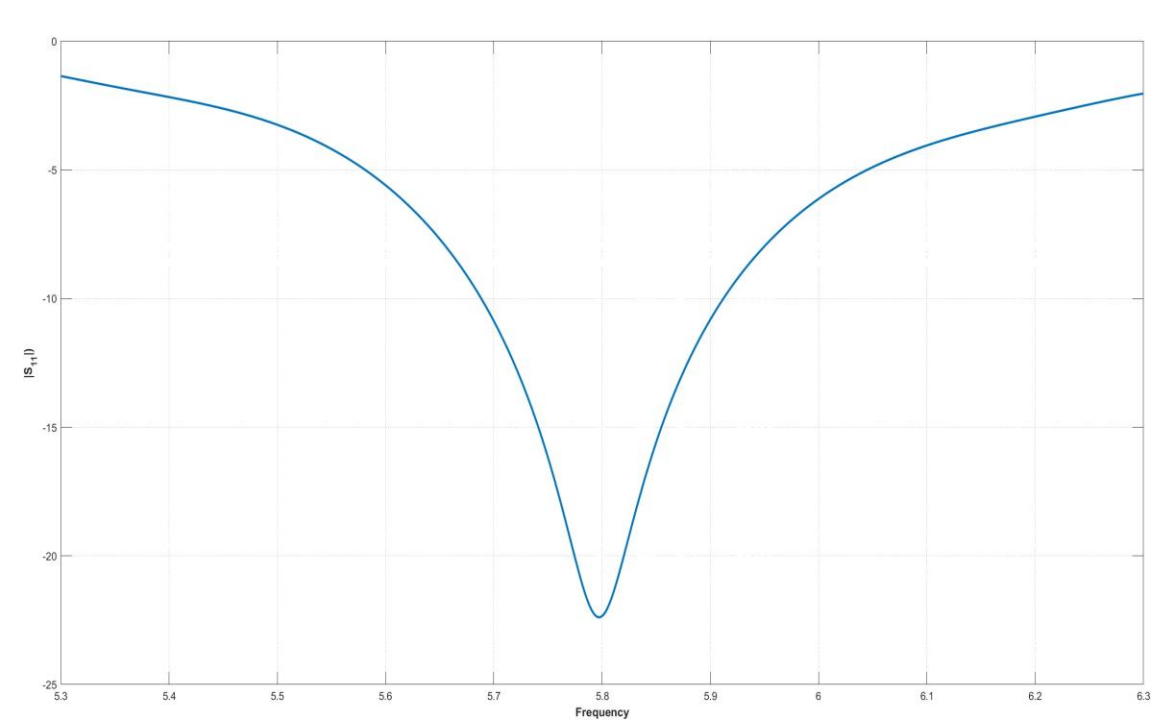


Fig 2: S11 – ideal substrate with no losses

Both the S11 plots show a resonance around 5.8 GHz, so the operating frequency is the same in both cases. The ideal substrate with no losses has a much deeper minimum (around -22 dB) compared to the real substrate (around -16 dB), this shows better impedance matching and lower reflection. The real substrate introduces additional losses and imperfections, that degrade the antenna performance.

Radiation patterns:

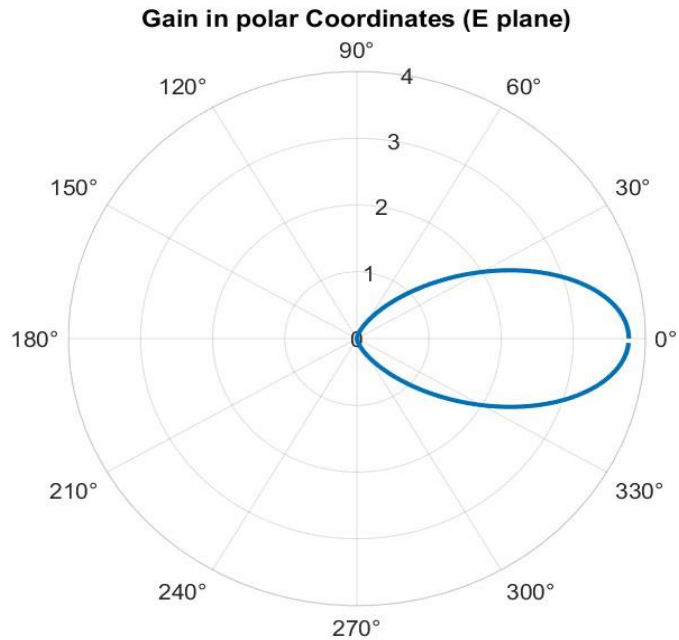


Fig 3: Gain in the E plane for real substrate

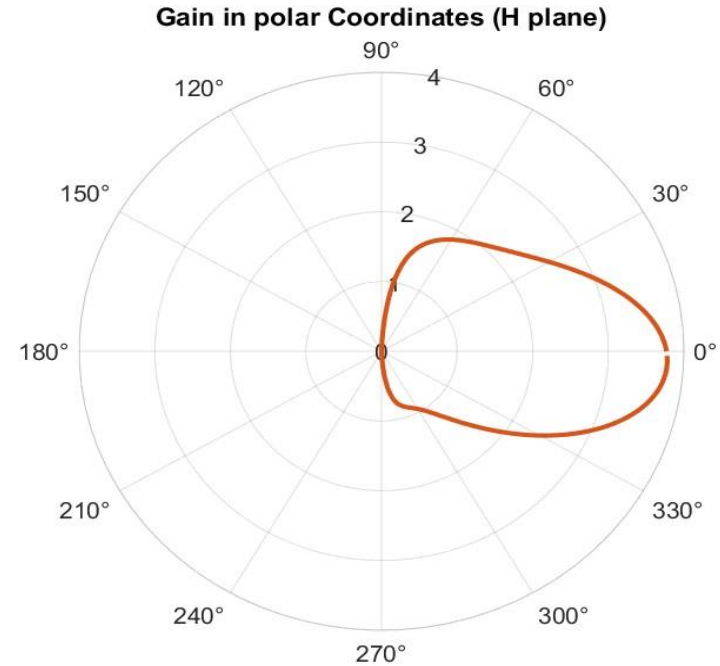


Fig 4: Gain in the H plane for real substrate

The CST patterns and homework plots agree in overall shape, both showing a broad broadside lobe with similar E- and H-plane behaviour. I used linear scale in my homework for the gains. The azimuthal plane shows a near-uniform arc in both cases, as expected for a patch. The slight asymmetry in my homework x-z plot comes from the simplified analytical model used, whereas CST gives a cleaner result due to its full-wave solver.

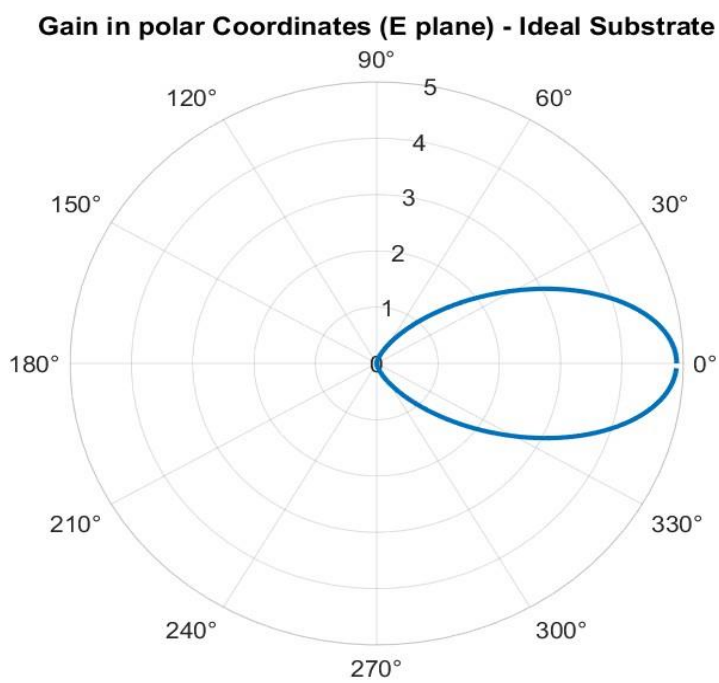


Fig 5: Gain in the E-plane in ideal substrate

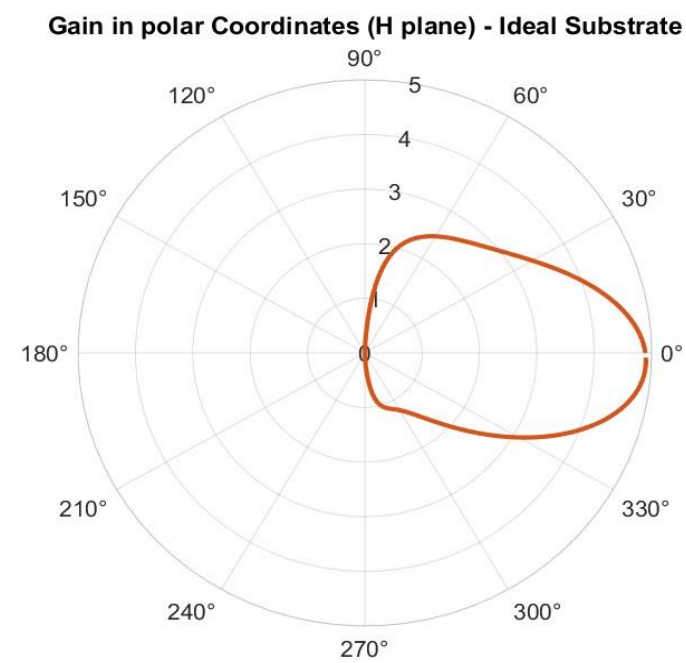


Fig 6: Gain in the H-plane in ideal substrate

Polarization and Max Gain: The polarization of the patch antenna is linear directional and the maximum gain in the direction of maximum radiation are:

Real substrate: Max Gain = 5.76dBi

Ideal substrate: Max Gain = 6.9dBi

I observed a reduction in gain in the real substrate case, from 6.9 dB in the ideal case to 5.76 dB, it could be attributed to dielectric losses in the substrate material, which absorb a portion of the input power as heat rather than contributing to radiation. Despite this, the overall radiation behaviour remains consistent between the two cases, with the main lobe directed broadside in both the E- and H-planes. The H-plane pattern reveals slight asymmetry and distortion in the real substrate case, highlighting the non-ideal material properties, though the performance degradation remains within acceptable limits for practical applications.

Part 2:

In Part 2 of the experiment, once the VNA was calibrated, we recorded how S_{11} of the antenna prototype changes across the same frequency range used in Part 1.

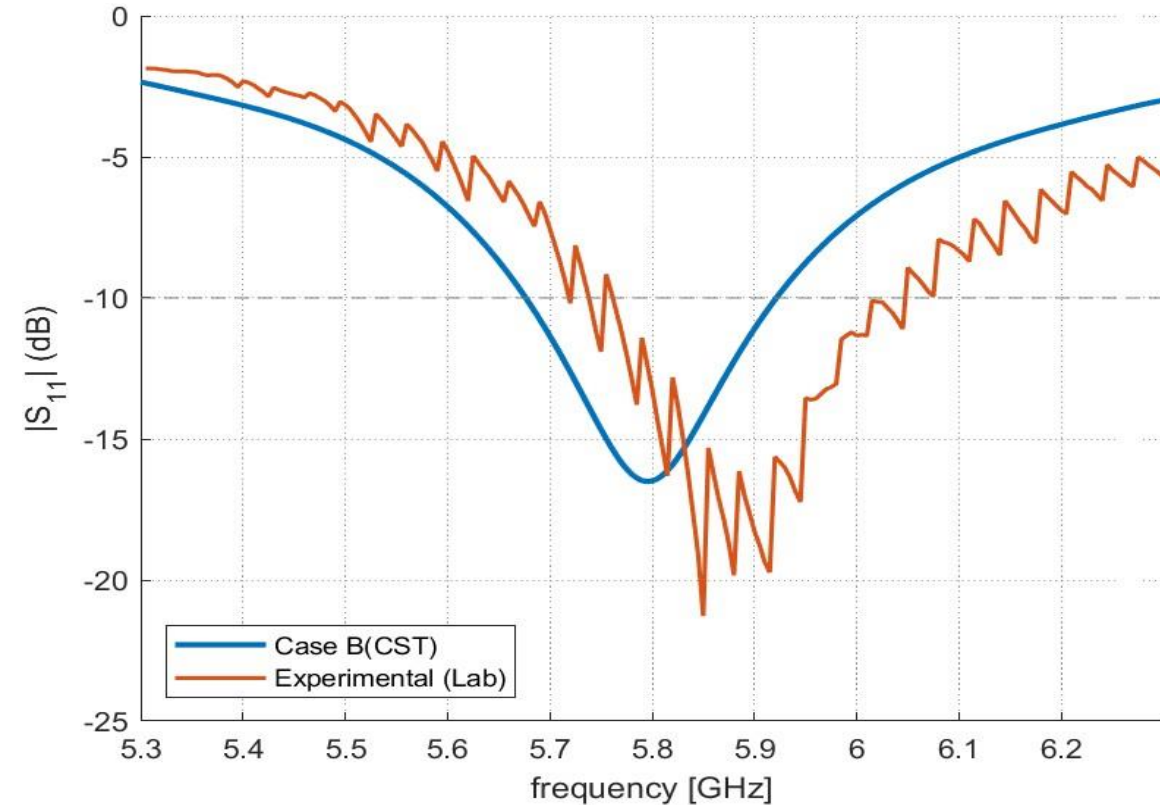


Fig 7: S_{11} comparison between experimental results and numerical simulation (real substrate)

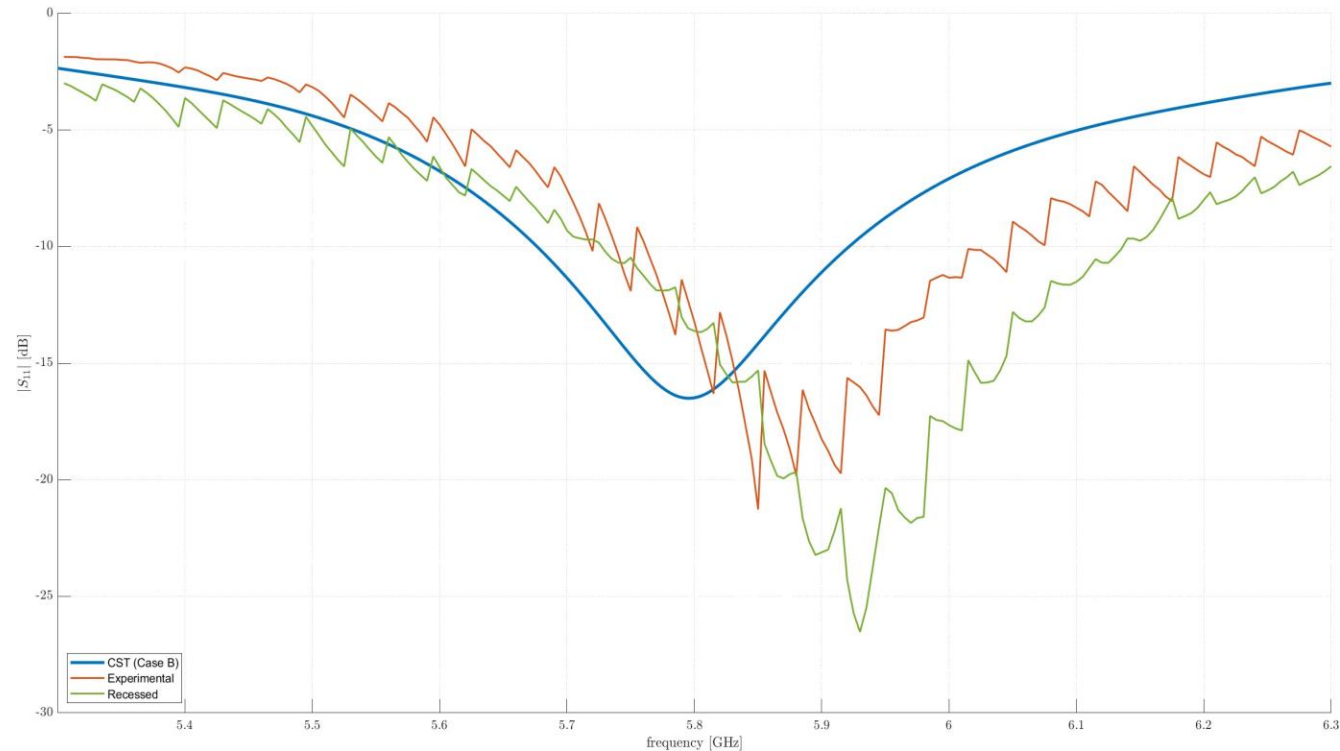


Fig 8: S_{11} in the case of insets

Conclusion:

The simulation and experimental results agree well in terms of resonant frequency, the measured S11 values closely follow the trend predicted by the simulation, The experimental curve exhibits significant ripple due to connector imperfections, cable effects, and residual calibration errors, but follows the same general trend as the CST result, validating the numerical model.

The inset-fed antenna achieves a deeper matching minimum of approximately -27 dB compared to -21 dB of the QWT, this indicates a better impedance matching at resonance. However, the heavy oscillations in the inset measurement makes it difficult to reliably determine its operational bandwidth, making the QWT response comparatively more reliable despite its shallower minimum.